

Sofa-so-good: Empowering Business Intelligence with Lakehouse Architecture

Subtitle: Transforming Retail Analytics through Scalable Cloud Architecture



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Introduction

Sofa-so-good is revolutionizing furniture retail by adopting a modern data strategy based on Lakehouse Architecture—a unified data solution that combines the flexibility of data lakes with the performance of data warehouses. In today’s competitive market, having streamlined, scalable, and efficient cloud architecture is essential for enabling data-driven decisions, and Sofa-so-good’s system delivers just that. By integrating structured and unstructured data from various business domains—such as sales, inventory, e-commerce, and web analytics—Sofa-so-good is unlocking the true potential of its data. This cloud-based framework ensures real-time insights, operational excellence, and future-ready scalability.

Why Lakehouse? Why It Works for Sofa-so-good

Why It Works:

- **Scalable & Adaptive:** It quickly expands with growing data volumes and new business needs.
- **Secure & Controlled:** It implements robust, role-based access control for enhanced governance.
- **Optimized & Efficient:** Ensures high performance while minimizing resource consumption.

Why Lakehouse Architecture?

- **Best of Both Worlds:** Stores vast unstructured data (like a data lake) and delivers fast, structured analytics (like a data warehouse).
- **Flexible Processing:** Supports both real-time and batch data ingestion.
- **Cost-Effective Growth:** Scales affordably and efficiently as the business grows.

Project Objective and Purpose

Objective:

To build a **robust and scalable data infrastructure** that enables Sofa-so-good to seamlessly access, analyze, and act upon business-critical data.

Purpose:

- Centralize and integrate diverse data sources.
- Automate data processing for better speed and accuracy.
- Enable efficient data retrieval for real-time analytics and reporting.
- Provide a unified “single source of truth” for business decision-making.

Mission and Goals

Mission:

To implement a **high-performance, reliable data engineering pipeline** that ensures Sofa-so-good’s data is accurate, consistent, and instantly accessible—laying a foundation for advanced analytics and operational improvement.

Key Goals:

- **Centralize Data:** Build a unified repository to facilitate informed decision-making.
- **Optimize Data Flows:** Streamline pipelines using automation and real-time processing.
- **Future-Proof Infrastructure:** Develop adaptable systems emphasizing efficiency and reliability.

Design Phase

The Lakehouse Architecture was selected to manage a variety of structured and unstructured data across multiple operational domains. This includes sales, warehouse activity, online shopping behavior, and web usage analytics.

Overview of the Lakehouse Architecture

The architecture is divided into **three primary layers**: data input, processing and storage, and data output and analytics.

a) Data Inputs

1. Sales Transactional Data

- Batch-ingested from point-of-sale systems.
- Format: CSV, SQL tables.
- Used for: sales analysis, forecasting, inventory, and finance.

2. E-Commerce Data

- Stream + batch ingestion from online stores.
- Format: JSON, relational databases.
- Used for: customer behavior tracking, checkout analysis, marketing.

3. Website Logs

- Real-time stream data from user activity.
- Format: Unstructured log files.
- Used for: engagement analytics, conversion optimization, A/B testing.

4. Warehouse Operational Data

- Batch-ingested stock and shipping data.
- Format: CSV, database tables.

- Used for: inventory control, order fulfillment, supply chain efficiency.

b) Data Outputs / Deliverables

1. Sales Performance Dashboards

- Real-time product and region-based sales metrics.
- Users: Sales & Management Teams.

2. Inventory Management Reports

- Stock levels, reorder tracking.
- Users: Operations Team.

3. Customer Segmentation Reports

- Purchase trends and personalized insights.
- Users: Marketing & Sales.

4. Sales Forecasting Reports

- Predictive trends and future projections.
- Users: Management & Finance.

5. Web Analytics Dashboards

- Engagement, traffic, and conversion rates.
- Users: Marketing & Web Dev Teams.

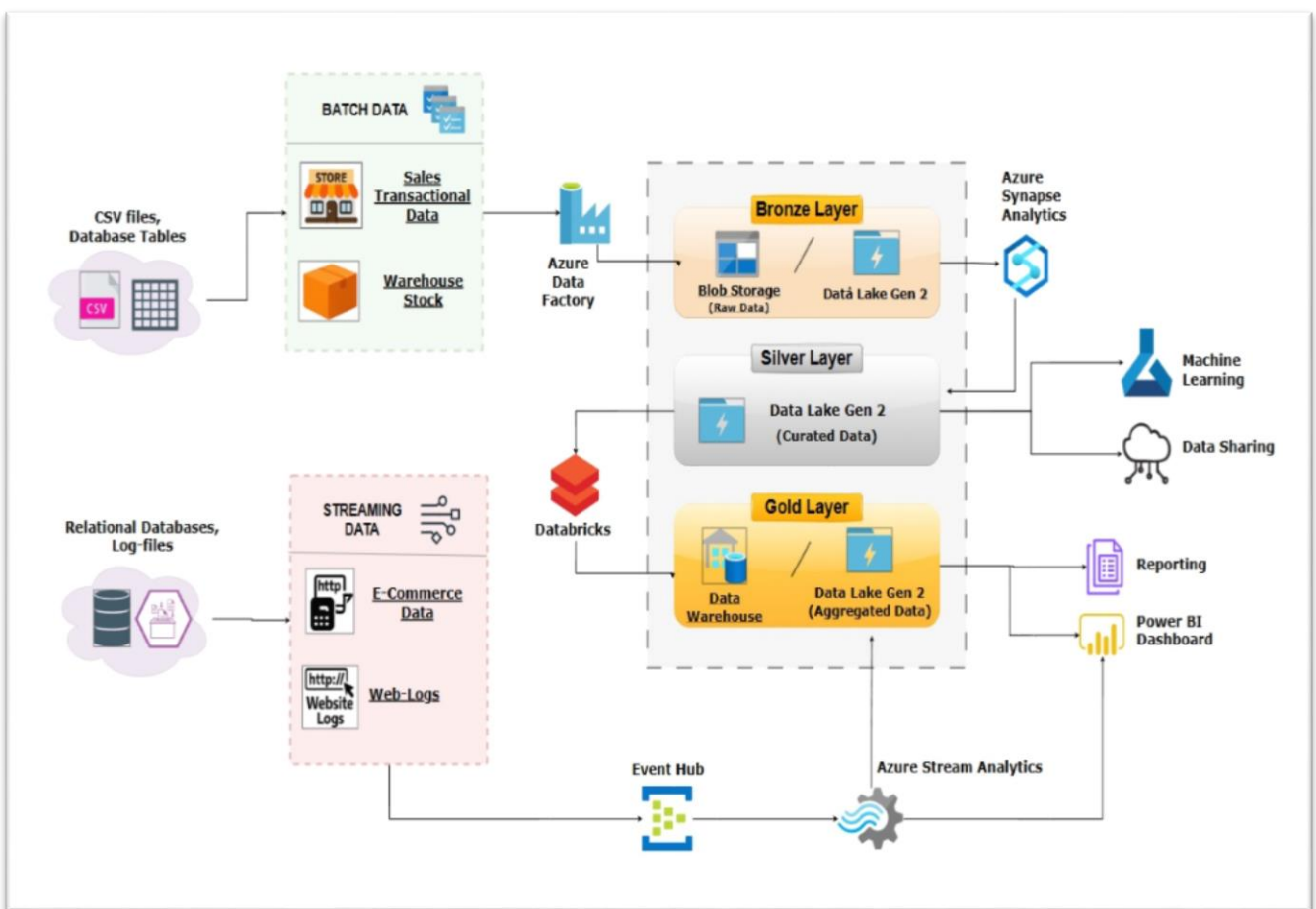
c) Data Users

- **Marketing Team:** Uses segmentation and web data for campaigns.
- **Sales Team:** Leverages dashboards and customer histories.
- **Operations Team:** Focuses on inventory optimization and logistics.
- **Management:** Depends on analytics for strategic planning.
- **Customer Service:** Accesses customer data for personalized support.
- **Web Development:** Analyzes user behavior to enhance site experience.

Cloud Architecture Phases

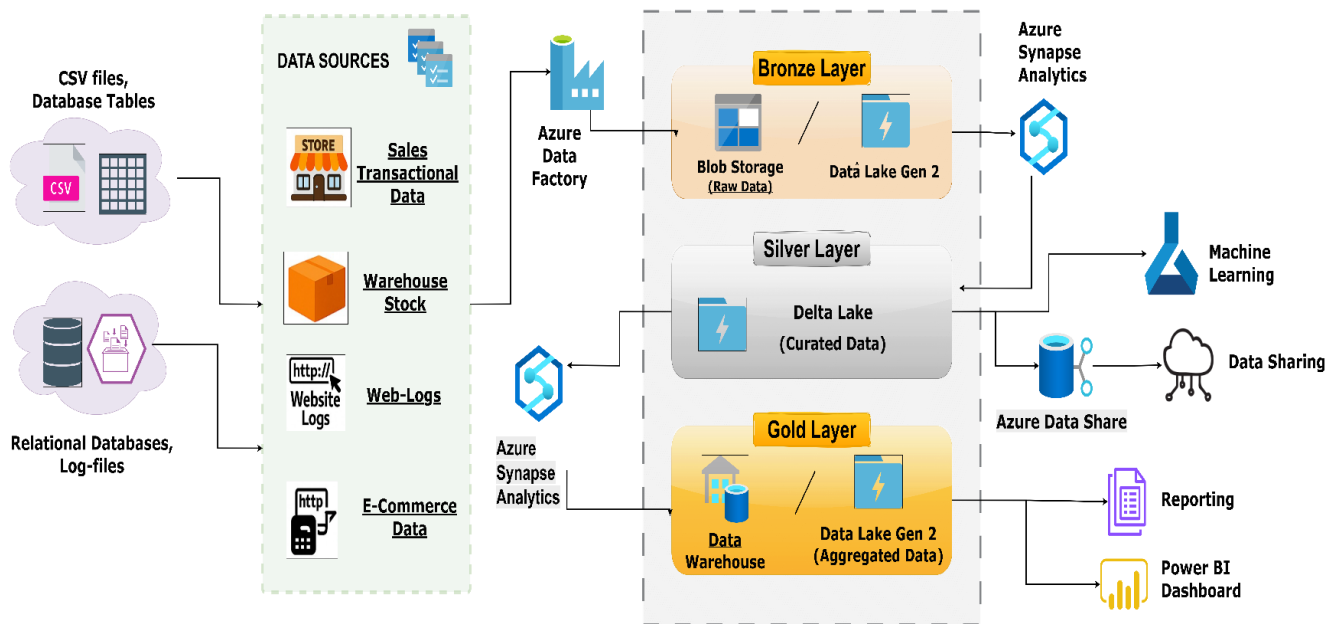
Phase 1: Data Ingestion and Storage

- Utilizes batch and stream pipelines via tools like Azure Data Factory or AWS Glue.
- Stores raw and curated data separately for compliance and operational use.
- Implements **Delta Lake** features like schema enforcement and ACID transactions.

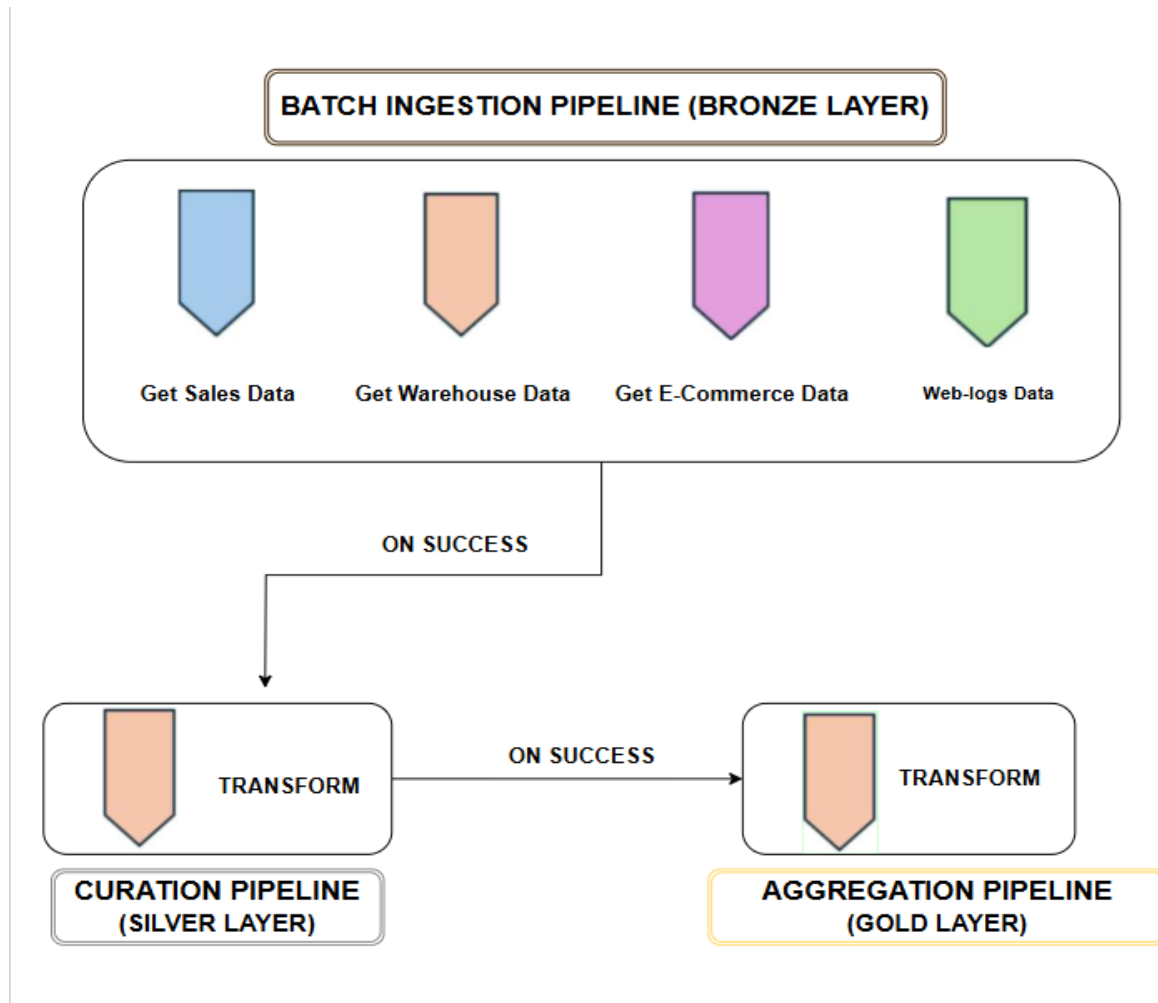


Phase 2: Data Processing and Analytics

- Batch and real-time processing through Spark or Databricks.
- Feeds dashboards and predictive models using Power BI, Looker, etc.
- Enables machine learning for sales predictions and customer targeting.



Master Pipeline flowchart



Conclusion

Sofa-so-good's Lakehouse-based cloud architecture is a forward-thinking solution that brings agility, accuracy, and intelligence to retail analytics. By harnessing the dual power of data lakes and warehouses, the company can unify its data landscape, scale confidently, and maintain a competitive edge in the evolving marketplace.

To conclude, our proposed cloud architecture is more than just a tech stack—it's a strategic initiative to help Sofa-so-good scale operations, improve decisions, and stay competitive through smart data usage.

This architecture not only takes into consideration technical development but is also a cost-effective and optimized solution according to business requirements.